Statistical approaches to estimate impact in small areas: a comparison with a focus on youth suicide prevention in rural counties

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LGG and CW conceived the study. LGG designed the analysis with VGR feedback. LGG implemented the analysis. LGG and HS drafted the manuscript. VGR and CW revised the manuscript critically and contributed to the interpretation of the results.

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Statistical approaches to estimate impact in small areas: a comparison with a focus on youth suicide prevention in rural counties

ABSTRACT

Estimating intervention effects in small areas is particularly challenging when outcomes are rare and data are sparse. In such settings, disease-mapping and small-area estimation models can improve precision by borrowing strength across space and time, yet their performance for causal impact evaluation under staggered adoption remains unclear. We compare alternative approaches for modeling untreated potential outcomes in county-level panels, including two-way fixed effects, cohort-based fixed effects, hierarchical Poisson models with random walk temporal trends, and a spatiotemporal specification combining autoregressive temporal structure with a scaled intrinsic conditional autoregressive spatial component.

Using a placebo simulation design that preserves the observed outcome process while resampling treatment timing under alternative assignment mechanisms, we evaluate each method's ability to recover cumulative effects in highly sparse data environments. Across data-generating processes, models incorporating structured temporal or spatiotemporal dependence yield modest but consistent gains in predictive accuracy at longer horizons relative to standard fixed-effects approaches. Flexible machine-learning methods leveraging outcome history show comparable performance.

A key finding is that pre-intervention predictive fit provides a practical diagnostic for counterfactual reliability. In simulations, post-period error increases sharply among counties with extreme pre-period misfit, and restricting attention to reliability-defined subsets materially alters substantive conclusions.

We apply these methods to evaluate the Garret Lee Smith Youth Suicide Prevention Program in rural U.S. counties using youth suicide mortality. Results underscore the importance of structured information sharing and predictive diagnostics when conducting impact evaluation in small-area, low-signal settings.

KEYWORDS

Impact evaluation, small area estimation, causal inference, youth suicide prevention, Bayesian disease mapping.

HIGHLIGHTS

- Compares impact estimators in sparse staggered-adoption panels
- Integrates small-area models with causal panel methods
- Uses placebo design with fixed outcomes and resampled assignment
- Introduces pre-fit diagnostic for counterfactual reliability
- Applies methods to youth suicide prevention in rural counties

1. Introduction

In the United States, suicide among youth has increased substantially in recent decades. Between 2001 and 2020, the suicide rate among individuals aged 10 to 24 rose from 6.9 to 10.5 per 100,000, representing a 52.2% increase (CDC, 2024). Increases have been particularly pronounced outside metropolitan counties—classified as micropolitan and noncore counties—where the suicide rate for this age group rose by 56.2% over the same period (CDC, 2024). Across metropolitan categories, the suicide rate among American Indian/Alaska Native youth has increased even more sharply, rising from 17.2 to 36.2 per 100,000 in rural areas, a 110.7% increase (CDC, 2024).

Federal efforts to address the ongoing suicide crisis in the United States began over two decades ago with the U.S. Surgeon General's Call to Action to Prevent Suicide (U.S. Public Health Service, 1999), followed by the passage of the Garrett Lee Smith Memorial Act (GLSMA) and the launch of the Substance Abuse and Mental Health Services Administration's (SAMHSA) GLS Youth Suicide Prevention Grant Program. To guide the development and implementation of such programs, rigorous evaluation of their impact is essential. However, impact evaluations in this context face important challenges. In particular, obtaining reliable community-level estimates of suicide rates can be difficult in small communities and among high-risk subpopulations because populations are small and suicide is a relatively rare event (Rossen et al., 2018). As a result, estimates for individual counties can be highly variable even when aggregate regional trends appear stable.

Bayesian disease-mapping (DM) models, which build upon methods from small area estimation (SAE), were developed to improve estimation of disease and other health outcome rates in small geographic areas by borrowing information across neighboring locations. In this study, we bring together these approaches with methods from causal inference for panel data to evaluate the impact of suicide prevention programs in small communities. While causal panel methods are widely used for program evaluation, they typically rely primarily on within-unit variation and may perform poorly when outcomes are sparse and noisy at the community level. By contrast, SAE and DM approaches explicitly leverage spatial and temporal structure to stabilize estimates through partial pooling or "information sharing."

Using both simulated and real data, this research examines how combining these frameworks can improve impact estimation in small areas. Specifically, we compare statistical approaches that integrate causal panel methods with information-sharing strategies from SAE and DM models to estimate counterfactual outcomes in data-sparse settings. Our empirical application focuses on the SAMHSA-funded Garrett Lee Smith (GLS) suicide prevention program and on

small, high-risk communities—such as tribal communities—that were underrepresented in previous evaluations (Walrath et al. 2015; Godoy Garraza et al. 2015, 2016, 2019). We compare alternative approaches in terms of their ability to estimate both the average program effect and the distribution of effects across counties, capturing potential heterogeneity in program impacts across small communities.

1.1 Motivating case

Mandated in the GLSMA, the GLS program provides support for states and Tribes to implement community-based suicide prevention and early intervention strategies for youth up to age 24. This program aims to increase the identification of at-risk youth, increase the capacity of clinical services providers, and to improve continuity of care after the identification of an at-risk youth or a discharge from an emergency room or inpatient treatment (SAMHSA, 2024). As of the start of the Federal Fiscal Year 2025, SAMHSA has funded 288 state and tribal grants to 50 states, two U.S. territories, and 74 tribes.

Initial studies on the impact of the GLS program on suicide mortality and morbidity among youth found evidence of a decrease in suicidal behavior rates among youth in counties implementing GLS-funded activities, compared to matched counties not implemented GLS activities (Walrath et al., 2015; Godoy Garraza et al., 2015). Later studies attributed long-term impact to prolonged exposure to GLS-funded activities (Godoy Garraza et al., 2019). A heightened effect on more rural counties was also reported. However, these studies did not include the most rural counties (termed "noncore" following the National Center for Health Statistics urban-rural classification scheme, see Ingram & Franco, 2014), which includes counties with the highest rates of suicide among youth and the largest increases between 2001 and 2020.

This study uses suicide mortality data from a sample of 516 counties across 10 states between 2007 and 2016, including data from 240 noncore counties—i.e., the smallest and most rural counties. Two-thirds of the counties in the sample had exposure to GLS suicide prevention programming during that period. Although more recent suicide prevention initiatives have been implemented since then, this dataset provides a well-characterized setting with consistent program exposure and outcome data, making it well suited for methodological comparison. This same sample of counties over the same period was previously examined in Godoy Garraza et al. (2021), which focuses on a different outcome as discussed further below. Revisiting the same setting allows us to compare how alternative modeling approaches perform when estimating program effects in small, data-sparse communities.

1.2. Related literature

This study lies at the intersection of two strands of research: (i) causal inference with panel data—particularly under staggered adoption—and (ii) small area estimation (SAE), including disease mapping (DM).

Impact evaluation using panel (or time-series cross-sectional) data has been an especially active area of research over the past two decades. Recent reviews such as Arkhangelsky & Imbens (2023) and Xu (2022) summarize key methodological developments. A growing body of work has emphasized the limitations of the widely used two-way fixed effects (TWFE) framework when moving beyond the canonical two-group, two-period setting to more general designs with staggered adoption and heterogeneous treatment effects (De Chaisemartin & D'Haultfœuille, 2020; Goodman-Bacon, 2021; Sun & Abraham, 2021). In response, several more flexible alternatives have been proposed. Borusyak et al. (2021) recast staggered-adoption difference-in-differences as an imputation problem, modeling the untreated potential outcome $Y_{it}(0)$ using pre-intervention data and imputing its missing values after adoption. Wooldridge (2023), extends this imputation logic to nonlinear settings, including generalized linear models suitable for discrete, nonnegative outcomes such as suicide counts, and shows that equivalent estimators can be implemented using cohort fixed effects rather than unit fixed effects. In this view, the central task becomes accurate modeling of the untreated outcome process—an objective that aligns naturally with the goals of small area estimation.

Small area estimation (SAE)—the estimation of quantities for subpopulations or domains where direct estimates are unreliable due to small sample sizes—and its related field of disease mapping (DM), which focuses on the geographic distribution of health outcomes, are well-established areas of research. Reviews of SAE and DM can be found in Pfeffermann (2013) and Waller & Carlin (2010). These approaches have been widely applied in contexts such as resource allocation, hotspot detection, and disease surveillance. More recently, their application to impact evaluation has received increasing attention (Kolak & Anselin, 2020; Papadogeorgou & Samanta, 2024; Reich et al., 2021; Schnell & Papadogeorgou, 2020).

A central feature of SAE and DM methods is the exploitation of spatial and/or temporal dependence across observational units to improve estimation through partial pooling or "information sharing." Such dependence can, however, give rise to interference or spillover effects, whereby exposure in one location affects outcomes in another (Kolak & Anselin, 2020). Spatial structure may also reflect unmeasured contextual factors. Some approaches leverage this structure explicitly for causal identification—for example, by jointly modeling outcome and exposure processes (Schnell & Papadogeorgou , 2020) or by assuming location acts as a proxy for latent confounders (Gilbert et al., 2024).

Our focus differs in two important respects. First, we consider a setting in which interference is assumed absent—an assumption we argue is plausible for a community-based prevention program targeting a noncommunicable health outcome. Second, rather than relying on spatial structure as a proxy for unmeasured confounding or jointly modeling exposure and outcome through shared spatial components, we adopt a different analytic strategy. Specifically, we model the relationship between latent factors and treatment assignment through an approach akin to a pattern-mixture formulation. This framework separates the role of spatiotemporal modeling from causal identification: spatial and temporal structure are used primarily to improve prediction of untreated potential outcomes in data-sparse settings, rather than serving as the identifying mechanism itself. By integrating causal panel-data methods with small-area estimation and disease-mapping techniques in this way, the approach allows structured information sharing to support counterfactual estimation while maintaining a transparent causal interpretation.

This study revisits the GLS setting analyzed in Godoy-Garraza et al. (2021), which examined suicide-related hospitalizations in small counties using disease mapping methods. Whereas that earlier work emphasized substantive program evaluation, this study centers on methodological comparison, using suicide mortality—an even more data-sparse outcome—as a case study to evaluate alternative modeling strategies.

2. Methods

2.1 Set up and notation

For a number of counties, indexed by $i = 1, \dots, N$, we observe an outcome of interest, Y_{it}, such as the number of youth suicides, over several years, indexed by $t = 1, \dots, T$. During this period, some counties are exposed to a "treatment", such as a suicide prevention program, starting in year $A_i \in \mathcal{A}$, $\mathcal{A} = \{a_1, \dots, a_J\}$. Once a county's exposure starts, it is considered to be exposed for the remainder of the study period (a setting known as staggered adoption). Counties not exposed during the entire study period are assigned $A_i \equiv \infty$. We define the binary indicator $W_{it} \equiv 1_{[A_i \leq t\,]}$, where $1_{[.]}$ is the indicator function, signaling that county i has been exposed to the intervention prior to, or at, time t. We denote by X_{it} the observed information available at time t for county i. In our core specifications, X_{it} consists primarily of the history of the outcome for the unit itself and, where applicable, neighboring units. Additional covariates (e.g., unemployment rates) that are plausibly unaffected by the intervention can also be incorporated; their inclusion is discussed in Supplementary Section S2.

2.2 Estimand

Using the potential outcome framework (Imbens & Rubin, 2015; Rubin, 1974), the impact of the intervention in each county and year following adoption is defined as the difference between the observed outcome and the outcome that would have been observed had county not been exposed:

$$\delta_{it} \equiv Y_{it} - Y_{it}(\infty), \tag{1}$$

for the subset of counties and years with indices (i, t) such that $W_{it} = 1$. The counterfactual outcome, $Y_{it}(\infty)$, is not directly observed and must be estimated using available information. Primarily, we use the history of the outcome, Y_{it} for the subset of counties and years with indices (i, t) such that $W_{it} = 0$. Additional auxiliary information such as other unit-level characteristics X_{it} —including population size and sociodemographic factors—can be utilized, provided they are either constant or vary independently of the intervention.

Interest generally lies in a summary measure of δ_{it} across all or a subset of post-intervention units and periods. Specifically, we focus on the cumulative effect up to four years following the start of the intervention among the smallest and most rural counties, denoted as "noncore" (nc):

$$\delta_s^{nc} = \frac{1}{N_{nc}} \sum_{i:i \in nc} \left(\sum_{t:A_i \le t \le A_i + s} \delta_{it} \right), \tag{2}$$

for $s \in [0, 4]$, where $N_{nc} = \sum_i 1_{[i \in nc]}$.

2.3 Assumptions

Estimating the impact of an intervention using observational data relies on several key assumptions.

First, we assume a stable intervention with no spillover effects (SUTVA; Rubin, 1974). This assumption can be problematic in spatial settings (Kolak & Anselin, 2020). Importantly, it does not preclude spatial or temporal dependence between potential outcomes. Such dependence may arise from shared underlying drivers—including factors that influence program assignment—without implying that exposure in one county directly affects outcomes in another. Our assumption rules out interference (e.g., benefits of the suicide prevention program

"spilling over" into untreated communities), while allowing for correlated potential outcomes across space and time.

Second, we assume that prior to intervention, observed outcomes coincide with the never-treated potential outcome.

Assumption 1 (No anticipation).

$$Y_{it}(\infty) = Y_{it}, \quad \text{for } t < A_i.$$

This is the standard no-anticipation condition in panel settings and is generally uncontroversial in our context.

Finally, because program adoption is not randomized, we allow for systematic differences between counties exposed at different times. Rather than assuming strict ignorability conditional on observed covariates alone, we posit a structured form of latent confounding.

Assumption 2 (Latent unconfoundedness).

There exists a (possibly multidimensional) latent factor U_i, with restricted structure, such that

$$\{Y_{it}(\infty)\}_t \perp A_i \mid \{X_{it}\}_t, U_i.$$

That is, conditional on observed covariates and a low-dimensional latent component capturing systematic differences across counties, adoption timing is independent of the entire path of untreated potential outcomes.

The key restriction is that U_i follows a structured form that can be learned from the data. The precise structure varies by model. For example, under TWFE, U_i corresponds to a latent time-invariant type captured by unit fixed effects; in spatiotemporal models, it may evolve smoothly over space or time according to a parametric dependence structure. Identification therefore relies not on the absence of unobserved heterogeneity, but on the assumption that such heterogeneity is sufficiently structured to be estimable.

2.4 Models to compare

All models specify the untreated outcome through a log-link count framework, with conditional mean

$$E(Y_{it}(\infty) \mid .) = n_{it} \exp(\eta_{it}). \tag{3}$$

where n_{it} is the known population at risk in county i at time t. In most specifications this corresponds to a Poisson model, which is standard for rare, non-negative count outcomes such as youth suicides.

Fixed effect models

A two-way fixed effect (TWFE) model is a frequent approach for the analysis of panel data (e.g., Angrist & Pischke, 2009). In its simplest form the model poses a unit-specific effects and time effects common to all counties (i.e., "parallel"),

$$\eta_{it} = \alpha_i + \gamma_t. \tag{4}$$

No distribution is assumed for the county-specific or the year-specific effects; county-specific effects may be related with the assignment A_i in any way, including spatially structured. Although the unit-specific effects are treated as nuisance parameters, average treatment effects can still be consistently estimated under standard assumptions (Martin, 2017; Wooldridge, 1999). Historically, the approach also involves effect homogeneity restrictions, say $\{\delta_{it}\}_{it} = \delta$, which have been the focus of more recent criticism and we avoid presently (De Chaisemartin & D'Haultfœuille, 2020; Goodman-Bacon, 2021; Sun & Abraham, 2021).

Wooldridge (2023) recently proposed a more parsimonious alternative, replacing unit-level by cohort-level fixed effects, i.e.,

$$\eta_{it} = \alpha_{A_i} + \gamma_t \tag{5}$$

Wooldridge (2023) shows that, under the exponential mean specification in (3) and in the absence of additional covariates, estimation of (5) yields numerically identical estimates of the common time effects γ_t and cohort-specific treatment parameters to those obtained from the unit fixed-effect specification in (4). When additional covariates are included, the two parameterizations generally produce different estimates. Moreover, the cohort-based formulation in (5) facilitates construction and inference for treatment effects on the outcome scale.

Mixed-effect, Temporal, and Spatiotemporal models

We consider three alternative models in the family of generalized linear mixed models of the type used for small area or domain estimation. The simplest alternative (RW), builds over the ETWFE by allowing cohort-specific trends,

$$\eta_{it} = \alpha_{A_i} + \gamma_t^{(A_i)}, \tag{6}$$

To flexibly model these secular trends, we assume that, for each cohort, the process $\left\{\gamma_t^{(a)}\right\}_{t=1}^{T}$ follows a random walk of order one. For $t > 1$ and for every a,

$$\gamma_t^{(a)} - \gamma_{t-1}^{(a)} = v_t^{(a)}, \qquad v_t^{(a)} \sim N(0, \tau_v^{-1}). \tag{7}$$

The trends are independent but share a smoothness parameter τ_v^{-1}, thereby allowing assignment-related heterogeneity to manifest through cohort-specific trend dynamics.

The subsequent model (RW-IID), retain the same flexibility, but further incorporate county-level departures from the cohort trends

$$\eta_{it} = \alpha_{A_i} + \gamma_t^{(A_i)} + \xi_i, \qquad \xi_i \sim N(0, \tau_\epsilon^{-1}). \tag{8}$$

The exchangeable county effect induces precision-based shrinkage: counties with sparse data are shrunk more strongly toward their cohort trend.

Our final model (RW-ST), rather than assuming exchangeable county effects, we model $\xi \equiv \{\xi_{it}\}$ as a Gaussian field with separable spatiotemporal covariance,

$$\xi \sim N\left(0,\ \tau_\epsilon^{-1}\ (R_S(\mathrm{g}) \otimes R_T)\right), \tag{9}$$

where g is the county adjacency graph, R_T is an AR(1) temporal correlation matrix with autocorrelation parameter ρ, and $R_S(\mathrm{g})$ is the reparameterization proposed by Simpson et al. (2017) of the well-known Besag–York–Mollié model (Besag et al., 1991), often referred as BYM2. Specifically, the spatial effect is expressed as a convex combination of a scaled intrinsic CAR component and an i.i.d. Gaussian component, with a mixing parameter $\phi \in [0, 1]$ controlling the proportion of variance attributable to spatial structure and a single precision parameter governing the overall marginal variance (Riebler et al., 2016). This specification induces spatial smoothing within years and temporal correlation within counties, allowing structured residual variation beyond cohort trends.

Gradient boosting

We approximate the log-risk surface using a gradient boosting ensemble (GB-Ens) with Poisson loss and log link. The predictor is constructed additively as

$$\eta_{it} = \sum_{m=1}^{M} \nu\ T_m(Z_{it}), \tag{10}$$

where each $T_m(\cdot)$ is a regression tree, ν is the learning rate, and trees are sequentially fit to the negative gradient of the Poisson deviance. This formulation accommodates nonlinearities and interactions while respecting the count nature of the outcome.

The feature vector Z_{it} is constructed from the observed information X_{it}, focusing exclusively on functions of untreated outcome histories. These include summaries of recent within-unit levels and trends, measures of sparsity in the count process, and aggregated information at the spatial and cohort levels (see Appendix A for details). Systematic differences across cohorts are captured through cohort-level historical summaries—such as cohort-specific averages of recent outcome levels and slopes among untreated units. These features allow the learner to adjust for persistent cohort heterogeneity while preserving flexibility in how such differences interact with unit-level dynamics. In this way, the model incorporates the structure required for identification without imposing additive fixed-effect restrictions.

In staggered-adoption panels, the set of observations revealing untreated outcomes depends on treatment timing. Rather than relying on a simple random train–validation split, model selection is tailored to the counterfactual objective. For each fit, we construct a perturbed adoption schedule that shifts treatment timing slightly earlier for eventually treated units (see Appendix B for details). Under this perturbed schedule, observations classified as "not yet treated" are used for training. Observations that become pseudo-treated under the perturbation—but remain untreated in the observed data—form a validation set. Because these units still reveal their untreated outcomes, they provide an out-of-sample proxy for evaluating predictive performance near treatment onset. The number of boosting iterations is chosen via early stopping by minimizing the Poisson negative log-likelihood on this pseudo-treated validation set. The stopping rule therefore targets predictive accuracy precisely in the region of the panel where counterfactual untreated outcomes must be imputed.

2.5 Estimation and Inference

Estimation of the untreated outcome model relies exclusively on observations with $W_{it} = 0$, that is, periods in which $Y_{it}(\infty)$ is observed. This reflects the imputation perspective adopted throughout: identification rests on modeling the marginal distribution of the untreated potential outcome process and predicting its missing values for treated periods.

We do not specify a joint model for untreated and treated potential outcomes. Instead, post-adoption outcomes are left unrestricted, and treatment effects are defined as deviations from predicted untreated outcomes. Formally, we focus on the marginal model for $Y_{it}(\infty)$, without imposing additional structure linking it to observed post-treatment outcomes.

Inference in all specifications proceeds by summarizing the posterior predictive distribution of the missing counterfactual outcomes relative to observed post-intervention outcomes. The difference across approaches lies in how this predictive distribution is obtained.

For the hierarchical models (RE, RW-IID, RW-ST), we adopt a parametric Bayesian approach based on a latent Gaussian model. Priors are placed on model parameters and latent components, and the posterior predictive distribution is obtained via integrated nested Laplace approximation (INLA, Rue et al., 2009) using the R-INLA package with default penalized complexity priors for hyperparameters governing random effects, temporal smoothness, spatiotemporal precision τ_ξ, temporal autocorrelation ρ, and spatial mixing ϕ (Simpson et al., 2017).

For the TWFE, ETWFE, and gradient boosting specifications, the predictive distribution is obtained using a Bayesian bootstrap (Rubin, 1981), clustered at the county level. Rather than specifying a parametric prior on model parameters, the bootstrap assigns random weights to counties, inducing a nonparametric prior on the empirical distribution of observations. In each replicate, the untreated model is re-estimated and counterfactual predictions recomputed. TWFE and ETWFE are estimated using the fixest package (Bergé, 2018), and boosting is implemented using xgboost with Poisson loss and log link (Chen & Guestrin, 2016). For boosting, resampling also affects early stopping based on pseudo-treated validation sets, so that bootstrap replication induces an ensemble of fitted boosted models.

3. Simulation (placebo) study

We assess the performance of the alternative approaches using a placebo simulation based on the observed data. Rather than generating fully synthetic outcomes, we keep the observed untreated outcome process fixed and simulate alternative treatment adoption schedules. For each replication r, we resample adoption times $A_i^{(r)}$ and construct the corresponding intervention indicator $W_{it}^{(r)} = 1[t \geq A_i^{(r)}]$.

Because the underlying outcome process is unchanged and no intervention effect is introduced, the true treatment effect under each simulated assignment is zero. The exercise therefore constitutes a placebo study: any estimated effect reflects predictive error in imputing untreated outcomes rather than recovery of a nonzero causal signal.

For each replication, models are estimated using observations classified as untreated under the simulated schedule ($W_{it}^{(r)} = 0$). Predictive performance is evaluated on observations that are treated under the simulated schedule ($W_{it}^{(r)} = 1$) but remain untreated in the observed data.

These observations reveal their untreated outcomes and therefore provide an out-of-sample assessment of how accurately each method imputes counterfactual untreated values.

Simulated adoption times are generated by assigning each unit–year a probability of adoption based on scenario-specific assignment weights and sampling adoption times accordingly. In all cases the outcome process is held fixed and only the assignment mechanism varies. This design isolates how alternative assumptions about treatment assignment interact with each model's structural restrictions on the untreated outcome process.

The three assignment mechanisms considered are summarized in Table 1.

Scenario	Assignment mechanism	Interpretation
Observed-history (MAR)	Adoption probability depends on functions of observed outcome history and covariates.	Approximately ignorable assignment conditional on observed information.
Latent level	Adoption probability depends on an unobserved unit-specific level component.	Persistent latent heterogeneity influences assignment.
Latent trend	Adoption probability depends on an unobserved unit-specific trend component.	Assignment correlated with latent outcome growth rates.

Table 1. Simulation assignment mechanisms. Full details of the assignment weight construction are provided in the Appendix C.

3.1 Performance measures

Predictive performance is evaluated using observations that are treated under the simulated assignment but remain untreated in the original data. These "recoverable" observations provide out-of-sample realizations of the untreated outcome against which counterfactual predictions can be compared.

For each replication, we compute performance measures over these validation observations and then average the resulting scores across replications. The main text focuses on the root mean squared error (RMSE), which summarizes point prediction accuracy.

In addition to RMSE, we assess predictive performance using several complementary measures that evaluate different aspects of the predictive distribution (Czado et al., 2009; Bracher et al., 2021). These include absolute error, Poisson deviance, coverage of nominal 95% prediction intervals, and proper scoring rules such as the logarithmic score and interval score. Proper

scoring rules jointly assess calibration and sharpness of the predictive distribution, rather than focusing solely on point accuracy. Detailed definitions of all performance measures are provided in Supplementary Section S1.

In all cases, predictions are evaluated against the observed untreated outcome, and interval estimates correspond to the 2.5% and 97.5% quantiles of the posterior predictive distribution.

3.2 Simulation results

To better understand how pre-intervention model fit relates to predictive performance, we examined post-period RMSE as a function of pre-period RMSE percentile in the placebo simulations (Figure 1). Within each replication and model, counties were ranked by their pre-period RMSE, and mean post-period RMSE was then computed across replications for each percentile. Across all three data-generating processes, post-period predictive error remains relatively stable across most of the pre-fit distribution. However, a pronounced increase in error appears in the extreme upper tail: counties in approximately the top decile of pre-period RMSE exhibit substantially higher post-period RMSE.

The deterioration is concentrated in this upper tail rather than gradual across percentiles, suggesting that extreme pre-period misfit identifies a subset of counties for which post-period prediction is unreliable. On the basis of this diagnostic result, subsequent performance comparisons focus on a reliability-restricted sample that excludes counties in the upper pre-fit tail. For completeness, results using the full placebo sample are reported in Supplementary Section S1, but primary comparisons are conducted within the reliability-defined subset.

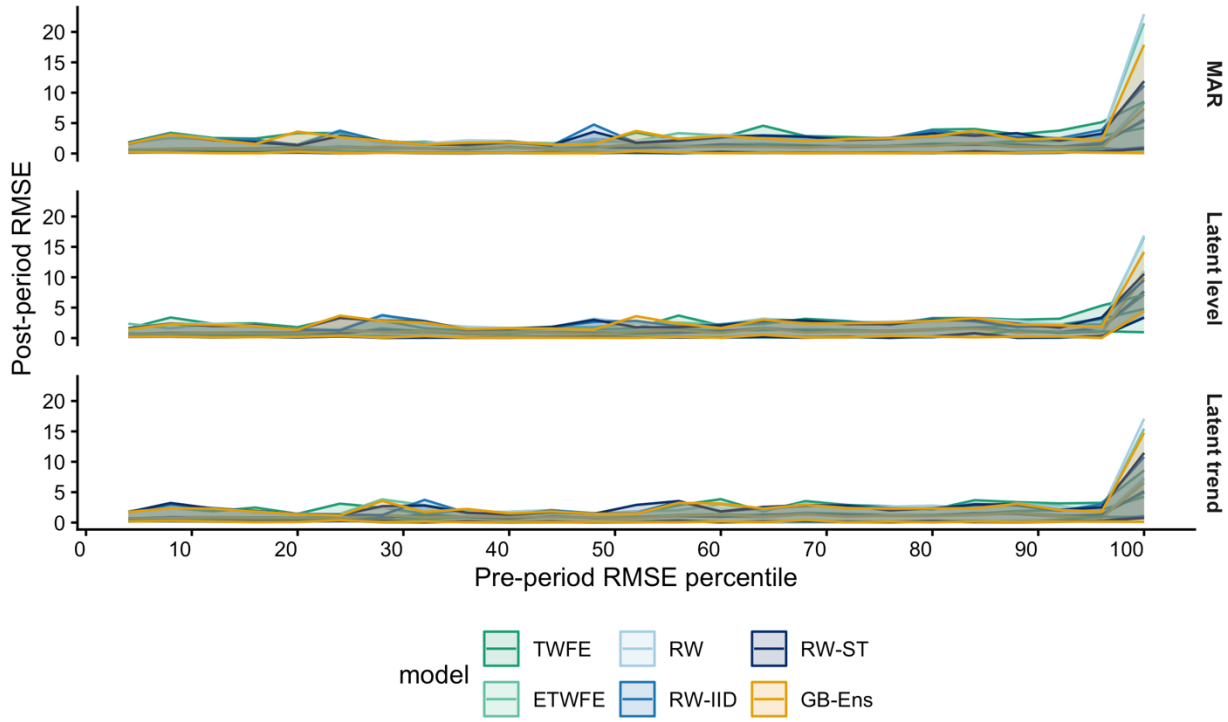


Figure 1 Relationship between pre-intervention model fit and post-period predictive performance in placebo simulations. The x-axis shows the percentile of county-level pre-period RMSE (computed within model and replication), and the y-axis shows the mean post-period RMSE across replications. Panels correspond to different data-generating processes (MAR, latent level, latent trend). Lines represent the average post-period RMSE across placebo replications; shaded bands indicate Monte Carlo uncertainty intervals. Across all data-generating processes and models, predictive performance remains relatively stable across most of the pre-fit distribution but deteriorates sharply in the upper tail (approximately the 90th–100th percentiles).

Figure 2 summarizes predictive performance in the placebo simulations for noncore counties, restricting attention to the reliability-defined sample. Overall, differences across models are modest, particularly at year 0. However, by year 4, models that leverage structured information—either through explicitly parameterized temporal or spatiotemporal components (RW-IID and RW-ST) or through flexible data-adaptive learning of outcome histories (GB-Ens)—tend to achieve slightly lower RMSE than TWFE and ETWFE across data-generating processes.

This pattern persists even under the latent-level assignment mechanism, which most closely aligns with the assumptions underlying two-way fixed effects models. In noncore counties, TWFE does not exhibit a systematic performance advantage under this assignment. By contrast, in core counties—where outcomes are less sparse and signal is stronger—TWFE performs comparatively better under the latent-level scenario (see Supplementary Section S1). These results suggest that in highly sparse settings, modest gains from either explicitly modeled

dependence structures or flexible history-based learning can persist even when the assignment mechanism favors simpler fixed-effect specifications. Results are largely unchanged when additional county-level covariates are incorporated; predictive rankings remain stable and only minor improvements are observed for specifications without unit fixed effects (Supplementary Section S2).

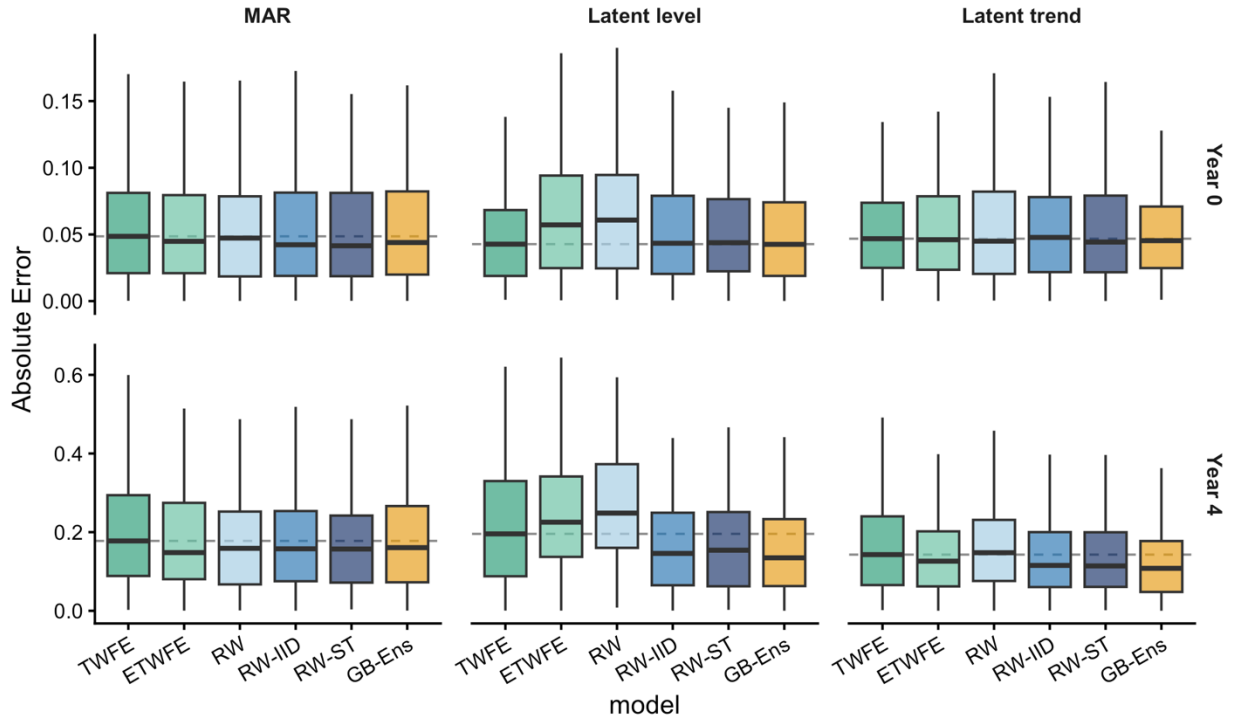


Figure 2 Predictive performance in placebo simulations among noncore counties, reliability-restricted sample (excluding the top 10% of counties by pre-period RMSE). Panels correspond to different data-generating processes (MAR, latent level, latent trend) and post-intervention horizons (year 0 and year 4). Performance is measured by root mean squared error (RMSE) of the estimated cumulative effect. Lower values indicate better predictive accuracy. Models shown are: two-way fixed effects (TWFE), enhanced two-way fixed effects (ETWFE), random walk (RW), random walk with independent and identically distributed county effects (RW-IID), random walk with spatiotemporal structure (RW-ST), and gradient boosting ensemble (GB-Ens). Boxes summarize performance across placebo replications.

4. Case study

4.1 Overall

Figure 3 contrasts cumulative effect estimates for noncore counties in the full sample and in the reliability-restricted sample (excluding the top 10% of counties by pre-period RMSE). In the full

sample, several models exhibit increasingly negative cumulative point estimates by years 3 and 4. However, these apparent iatrogenic patterns are not consistent across models and arise primarily when including counties identified in the simulation study as having poor pre-intervention fit and degraded predictive performance.

Once attention is restricted to the reliability-defined subset, the pattern changes materially. Across models, cumulative effects are centered close to zero in the early post-intervention years and remain modest in magnitude through year 4. The pronounced negative drift observed in the full sample is substantially attenuated. Given the placebo evidence linking extreme pre-period misfit to unreliable post-period prediction, the reliability-restricted results provide a more credible assessment of program impact in noncore counties. Within this estimable subset, we find no consistent evidence of large iatrogenic effects.

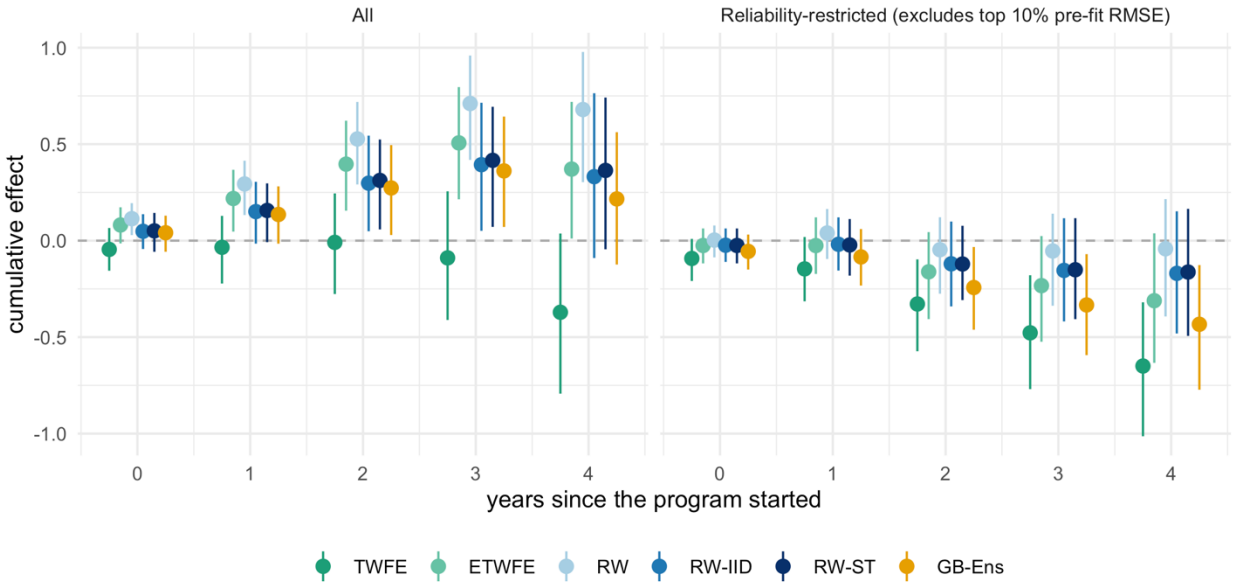


Figure 3 Estimated cumulative program effects in noncore counties as a function of years since program initiation. The left panel shows results for the full sample; the right panel restricts to the reliability-defined subset (excluding the top 10% of counties ranked by pre-period RMSE). Points represent posterior mean cumulative effects and vertical lines denote 95% credible intervals. Negative values indicate fewer observed deaths than expected under the estimated counterfactual (i.e., a reduction), while positive values indicate an increase relative to the counterfactual. Models shown are: two-way fixed effects (TWFE), enhanced two-way fixed effects (ETWFE), random walk (RW), random walk with independent and identically distributed county effects (RW-IID), random walk with spatiotemporal structure (RW-ST), and gradient boosting ensemble (GB-Ens).

4.2 Effect heterogeneity

Figure 4 summarizes heterogeneity in estimated cumulative effects across noncore counties, measured by the interquartile range (IQR) of county-level effects. In the full sample, the IQR increases steadily with years since program initiation, indicating growing dispersion in estimated impacts across counties over time. This pattern is observed across modeling approaches, with broadly comparable magnitudes by years 3 and 4.

Importantly, restricting the analysis to the reliability-defined subset does not materially reduce the degree of heterogeneity. The IQR remains of similar magnitude and continues to increase with exposure duration. This indicates that the observed heterogeneity is not driven by the 10% of counties identified as having poor pre-intervention fit. Rather, dispersion in estimated effects persists among counties for which counterfactual prediction is empirically supported by the placebo diagnostics. Thus, while the apparent iatrogenic pattern in the full sample is attenuated under the reliability restriction, meaningful cross-county variation in estimated program effects remains.

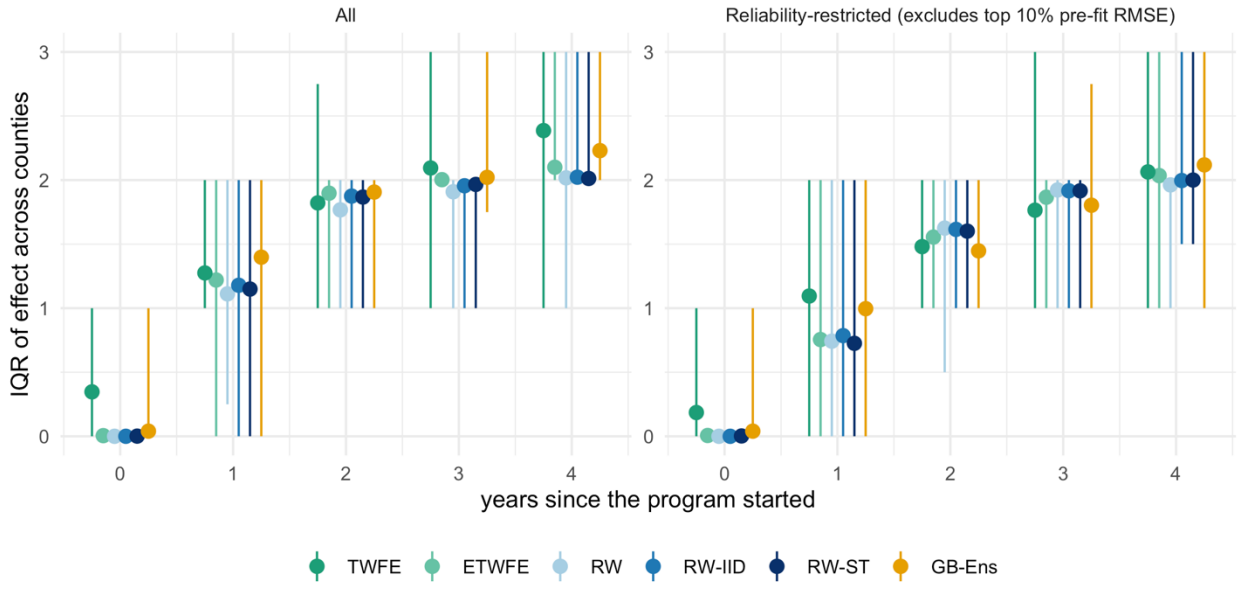


Figure 4 Interquartile range (IQR) of county-level cumulative effect estimates in noncore counties as a function of years since program initiation. The left panel shows results for the full sample; the right panel restricts to the reliability-defined subset (excluding the top 10% of counties by pre-period RMSE). Points represent the median IQR across posterior draws, and vertical lines denote 95% uncertainty intervals. Higher values indicate greater heterogeneity in estimated program effects across counties. Models include TWFE, ETWFE, RW, RW-IID, RW-ST, and GB-Ens.

Having established evidence of heterogeneity in cumulative effects across counties, we next examined whether this variation is associated with differences in program orientation. Godoy

Garraza et al. (2021) reported larger reductions in suicide-related hospitalizations in counties where a greater proportion of trainings were classified as "long." Longer trainings are typically directed toward health professionals rather than lay gatekeepers and may reflect an emphasis on strengthening treatment capacity rather than early identification. Motivated by this finding, we assessed whether the proportion of long trainings delivered in a county is associated with the magnitude of the estimated cumulative effect.

Table A2 reports the Spearman correlation between the proportion of long trainings and the estimated cumulative effect, stratified by years since program initiation and modeling approach. Although point estimates vary in sign and magnitude across years and models, a substantial portion of the posterior distribution overlaps zero in all cases. This pattern is consistent in both the full and reliability-restricted samples. We therefore find no systematic evidence that cross-county heterogeneity in estimated program effects is explained by differences in training composition.

5. Discussion

Estimating the effects of interventions in small areas—such as assessing a prevention program's impact on youth suicides in small rural communities—is inherently challenging due to sparse data and limited signal. Techniques developed for small area estimation (SAE), traditionally used for descriptive or surveillance purposes, can be adapted to estimate counterfactual outcomes in this context. In this study, we examined several models that incorporate shrinkage or structured "information sharing," alongside traditional fixed-effects approaches and data-adaptive models that do not rely on explicit random-effects structure. Overall, performance differences were not large in magnitude in the most data-constrained setting (noncore counties), but they were consistent in direction. Across data-generating processes, models that incorporate structured temporal or spatial components (RW-IID, RW-ST) or flexibly leverage outcome and neighborhood history (GB-Ens) tended to achieve lower RMSE at longer horizons. These gains were modest but systematic. The advantage of structured or flexible information sharing appears most relevant precisely in the sparse settings that motivate small area estimation. In larger (core) counties, where outcome counts are less sparse, performance differences across models were generally smaller, consistent with the expectation that when signal is stronger, simpler specifications can perform comparably well.

A central methodological contribution of this study is the demonstration that pre-intervention fit provides a practical diagnostic for post-intervention predictive reliability. In placebo simulations, post-period prediction error increased sharply among counties in the extreme upper tail of pre-period RMSE, indicating limited recoverability of counterfactual outcomes in

these units. Restricting attention to reliability-defined counties materially altered substantive conclusions in the case study: apparent late iatrogenic effects in the full sample were substantially attenuated once counties with extreme pre-period misfit were excluded. Importantly, evidence of heterogeneity in cumulative effects persisted under this restriction, indicating that cross-county variation is not merely an artifact of unreliable counterfactual estimation.

The application illustrates that model choice can meaningfully affect substantive conclusions in sparse settings. While some approaches suggested decreasing youth suicide rates in noncore counties four years after GLS implementation, effect estimates varied across models and remained sensitive to the reliability of counterfactual recovery. Strong evidence of heterogeneity across counties was observed regardless of modeling approach. However, the hypothesized mechanism—differences in program orientation as proxied by the proportion of longer trainings directed toward health professionals—showed little consistent association with effect magnitude. Thus, while heterogeneity appears robust, its drivers remain unclear, and the mechanism of action underlying potential program impacts warrants further investigation.

Several limitations should be acknowledged. First, the methodological findings arise in an exceptionally low-signal environment; conclusions may differ in settings with more abundant outcome data. Indeed, we observed clearer performance differences across models in larger counties. Second, causal interpretation relies on the assumption that, after conditioning on observed covariates, outcome history, and structured unit- and time-level components, no additional unobserved time-varying shocks differentially affecting treated counties remain. While the models considered allow for persistent unobserved differences (e.g., unit fixed effects) and structured temporal variation (e.g., cohort trends or autoregressive components), they do not protect against idiosyncratic time-varying confounders that are not captured by historical patterns. Such shocks could bias effect estimates, particularly at longer follow-up horizons.

In summary, our findings suggest that SAE techniques can be integrated into impact evaluations in small-area contexts, but careful attention to predictive diagnostics is essential. Pre-intervention fit can serve as a transparent and empirically grounded criterion for assessing the credibility of counterfactual estimates. Future research should explore hybrid modeling approaches—such as combining structured shrinkage with flexible machine learning methods—and further investigate the sources of cross-area heterogeneity in low-signal environments.

Data availability

The county-level mortality data analyzed in this study originate from the Centers for Disease Control and Prevention (CDC) but include restricted small-area counts that are not publicly available due to disclosure risk. The analytic dataset used in this study cannot be shared.

All analysis code required to reproduce the modeling, simulation, and figures is publicly available at: https://github.com/Lucas-GG/small-area-impact-estimation

The repository includes a synthetic ("mock") dataset that preserves the panel structure and variable definitions necessary to execute the full analysis pipeline without access to restricted data. Researchers interested in accessing restricted mortality data should consult the CDC regarding applicable data access procedures and data use agreements.

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Appendices

Appendix A: Feature expansion

To facilitate a meaningful comparison between the gradient boosting ensemble and the parametric panel-data models considered in this study, we construct a set of predictors derived from the history of the untreated potential outcome $Y_{it}(0)$. These predictors summarize recent levels, trends, and spatial and cohort-level patterns in the outcome process and are used exclusively in the gradient boosting model.

Within-county history

The average value of the outcome for county i over the last L periods is

$$\bar{Y}_{it}^{(L)}(0) \equiv L^{-1} \sum_{\ell} Y_{it-\ell}\,(0),$$

and the corresponding local slope is

$$\dot{Y}_{it}^{(L)}(0) \equiv \frac{\sum_{\ell}\left(\ell - \bar{\ell}\right)\left(Y_{it-\ell}(0) - \bar{Y}_{it}(0)\right)}{\sum_{\ell}\left(\ell - \bar{\ell}\right)^2}, \qquad \bar{\ell} = \frac{L+1}{2}.$$

The share of zero outcomes is

$$\bar{C}_{it}^{(L)}(0) \equiv L^{-1} \sum_{\ell} 1(Y_{it-\ell}(0) = 0).$$

To capture both short-run and medium-run dynamics in a sparse count process, we include each historical summary using $L \in \{3,7\}$. We also include the most recent outcome as an additional predictor $Y_{it-1}(0)$.

Spatial aggregation

Let G denote the binary contiguity (adjacency) matrix, with $g_{ij} = 1$ if areas i and j share a border and $g_{ii} = 0$. Let

$$\mathcal{O}_t \equiv \{i : W_{it} = 0\},$$

Denote the set of units with observed $Y_{it}(0)$ at time t. Define the available-neighbor degree

$$d_{it} \equiv \sum_{j \neq i} g_{ij} 1(j \in \mathcal{O}_t),$$

and corresponding time-varying row-standardized weight

$$\tilde{g}_{ij,t} = \begin{cases} \dfrac{g_{ij} 1(j \in \mathcal{O}_t)}{d_{it}}, & d_{it} > 0 \\ 0, & d_{it} = 0 \end{cases}$$

The neighbor averages of the level and slope are then

$$\bar{Y}_{it}^{(GL)}(0) \equiv \sum_{j \neq i} \tilde{g}_{ij,t} \bar{Y}_{it}^{(L)}(0), \qquad \dot{Y}_{it}^{(GL)}(0) \equiv \sum_{j \neq i} \tilde{g}_{ij,t} \dot{Y}_{jt}^{(L)}(0).$$

Cohort-level averages

Recall that A_i denote the assignment time or cohort of unit i. Define the number of untreated units in cohort a at time t as

$$N_{a,t}^{(0)} = \sum_i 1(A_i = a) 1(W_{it} = 0)$$

The cohort-level averages,

$$\bar{Y}_t^{(aL)}(0) \equiv \frac{1}{N_{a,t}^{(0)}} \sum_{i: A_i = a, W_{it} = 0} \bar{Y}_{it}^{(L)}(0), \qquad \dot{Y}_t^{(aL)}(0) \equiv \frac{1}{N_{a,t}^{(0)}} \sum_{i: A_i = a, W_{it} = 0} \dot{Y}_{jt}^{(L)}(0),$$

Contemporary averages

Let

$$N_t^{(0)} = \sum_i 1(W_{it} = 0),$$

denote the number of untreated units at time t. The contemporaneous control averages are

$$\bar{Y}_{\cdot t}(0) \equiv \frac{1}{N_t^{(0)}} \sum_{i: W_{it} = 0} Y_{it}(0).$$

Appendix B: Sticky reassignment of adoption times

To construct alternative but plausible treatment adoption schedules in a staggered-adoption panel, we simulate reassigned adoption times $A_i^{(r)}$ for each unit i. The procedure is designed to be sticky—i.e., simulated adoption times remain close to the observed adoption time and, in our implementation, are restricted to shift earlier (backward in time), which preserves the general adoption structure while creating variation in the not-yet-treated partition.

Step 1: Estimate adoption hazard

When hazard weighting is used, we first estimate a unit–year adoption propensity among not-yet-treated observations. Let

$$h_{it} \equiv \Pr(A_i = t \mid A_i \geq t, X_{it}),$$

where X_{it} includes functions of the unit's observed history (including lagged outcomes and derived summaries). We estimate h_{it} using a gradient boosting classifier and denote the fitted hazard by $\hat{h}_{it}$. (In the implementation, $\hat{h}_{it}$is stored as cpr.)

Step 2: Construct candidate adoption years

Let A_i be the observed adoption year for unit i. For never-treated units, we set $A_i^{(r)} = \infty$ in all replications. For treated units, we define a backward-only candidate set

$$\mathcal{C}_i = \{A_i - s, A_i - s + 1, \ldots, A_i\},$$

where s is a maximum backward window (in our implementation, $s = \max\{6, 2 \cdot \text{shift}\}$). Candidate years are restricted to the support of the observed panel years; when the panel has gaps, candidate years are snapped to the nearest observed year in the data.

Step 3: Define sticky (distance) weights

To keep simulated adoption times close to the observed A_i, we assign higher probability to candidates near A_i. For each candidate year $c \in \mathcal{C}_i$, define the backward distance $d = A_i - c \geq 0$. Sticky weights are proportional to a Poisson probability mass function:

$$w_i^{\text{sticky}}(c) \propto \Pr(D = d) = \text{PoissonPMF}(d; \lambda),$$

with λ = shift (so the typical backward shift is about one year when shift = 1).

A deterministic variant is also available, setting $A_i^{(r)} = A_i -$ shift, subject to the same clamping/snapping rules.

Step 4: Incorporate hazard weighting (optional)

To favor earlier shifts in unit–years with higher adoption propensity, we optionally multiply the sticky weights by a hazard term:

$$w_i^{\text{haz}}(c) \propto \left(\max\{\hat{h}_{i,c}, \varepsilon\}\right)^{\kappa},$$

where κ (hazard power) controls the influence of the hazard and $\varepsilon > 0$ prevents zero weights.

The final sampling weights are

$$w_i(c) \propto w_i^{\text{sticky}}(c)\, w_i^{\text{haz}}(c), c \in \mathcal{C}_i,$$

renormalized to sum to one.

Step 5: Sample reassigned adoption time and define treatment

For each treated unit i, we draw

$$A_i^{(r)} \sim \text{Categorical}(\mathcal{C}_i; w_i(\cdot)),$$

and define the simulated treatment indicator

$$W_{it}^{(r)} = \mathbf{1}\{t \geq A_i^{(r)}\}.$$

This simulated schedule is used to (i) define not-yet-treated observations for training and (ii) construct pseudo-treated validation observations that remain untreated under the observed schedule, thereby supporting counterfactual-oriented early stopping in the gradient boosting model.

Appendix C: Simulation assignment mechanism

To evaluate estimator performance under alternative assignment structures, we generate simulated treatment adoption schedules while holding the untreated outcome process $Y_{it}(0)$ fixed at its observed values. Each simulated "world" differs only in how treatment adoption times are generated. The outcome process itself is not regenerated.

Let A_i denote the observed adoption time in the data. For each simulated world and replication r, we generate new adoption times $A_i^{(r)}$ using unit–year–specific assignment weights. The treatment indicator is then defined as

$$W_{it}^{(r)} = \mathbf{1}\{t \geq A_i^{(r)}\}.$$

The procedure used to generate $A_i^{(r)}$ is common across worlds; only the construction of the assignment weights differs.

General reassignment procedure

For each unit–year (i, t), let $c_{it} \geq 0$ denote an assignment weight (interpretable as a hazard or propensity for adoption in year t). For each unit i, we sample an adoption time from the support of observed panel years with probability proportional to these weights:

$$\Pr(A_i^{(r)} = t) \propto c_{it}.$$

Thus, simulated adoption times are drawn from a unit-specific discrete distribution over calendar years. Unlike the sticky reassignment procedure used internally in the learning algorithm, the simulation does not restrict adoption to remain close to the observed start year. In particular, units observed to be never treated in the data may become treated in simulated worlds if their assignment weights are positive.

Because untreated outcomes are only observed prior to the realized adoption time in the data, simulated assignments may create additional missing potential outcomes. However, some observations that are post-treatment under the simulated schedule but pre-treatment under the observed schedule remain observed. These "recoverable" observations provide out-of-bag validation points for evaluating predictive performance.

World 1: Observed-hazard assignment (approximately ignorable)

In the baseline world, assignment weights are based on an estimated adoption hazard conditional on observed predictors. Specifically, we estimate

$$h_{it} = \Pr(A_i = t \mid A_i \geq t, X_{it}),$$

where X_{it} includes functions of historical outcomes and covariates. The hazard is estimated using gradient boosting, yielding fitted values $\hat{h}_{it}$. We then set

$$c_{it} = \hat{h}_{it}.$$

In this world, treatment timing is approximately ignorable conditional on the observed history and covariates used to estimate the hazard.

World 2: Latent level confounding

To generate assignment driven by persistent unobserved heterogeneity, we estimate a Poisson generalized linear mixed model with a random intercept:

$$Y_{it}(0) \sim \text{Poisson}(n_{it}\lambda_{it}), \log \lambda_{it} = \alpha + u_i,$$

with offset $\log n_{it}$. Let $\hat{u}_i$ denote the estimated unit-specific random intercept. Assignment weights are then defined as

$$c_{it} = \exp(\hat{u}_i),$$

repeated across years for unit i.

Under this world, treatment timing depends on a persistent latent "level" component that is not fully recoverable from finite-history summaries. Assignment is therefore ignorable conditional on the latent u_i, but not necessarily ignorable given observed history alone.

The random intercepts are estimated using all observed untreated outcomes, including years that would be unavailable under simulated assignments. This is acceptable because the latent effects are used solely to generate alternative assignment mechanisms and are not supplied to the learners.

World 3: Latent trend confounding

To generate assignment driven by unit-specific trends, we estimate a Poisson mixed model with a random slope:

$$\log \lambda_{it} = \alpha + u_i + s_i \cdot t,$$

where s_i represents a unit-specific trend component. Let $\hat{s}_i$ denote the estimated random slope. Assignment weights are defined as

$$c_{it} = \exp(\hat{s}_i),$$

again repeated across years for unit i.

In this world, adoption depends on latent growth rates rather than persistent levels. Assignment is therefore confounded by unobserved trend heterogeneity not fully captured by short-run historical features.

Replications

For each world, we generate multiple (200) independent replications of the assignment mechanism. Performance metrics are aggregated across replications to reduce Monte Carlo variability, particularly because the fraction of recoverable observations may be modest in any single draw.

Design principle

In all simulated worlds, the untreated outcome process $Y_{it}(0)$ is held fixed at its observed values. Only the assignment mechanism is resampled.

This design deliberately avoids imposing a fully synthetic outcome data-generating process. Instead, the empirical outcome process is held fixed and only the treatment assignment mechanism is varied. This approach isolates how alternative assumptions about treatment assignment interact with each model's restrictions on the untreated outcome process, while preserving the sparsity and distributional features of the real data.

Supplemental Material

S1. Additional Simulation Results

This appendix provides additional details on the placebo simulation study and supplementary results that support the findings reported in the main text. Firstly, we formally define the performance measures, all computed at the same aggregation level as the estimand in Section 2.2. Secondly, we present additional results, including performance among core counties and results based on the full sample (i.e., without restricting to the trusted subset that excludes the upper 10% of pre-treatment prediction error).

1. Performance Measures

This section defines the performance measures used in the placebo study. All measures are aligned with the estimand defined in Section 2.2 and are computed at the subgroup and event-time level.

Aggregated counterfactual quantities

Recall that the estimand of interest is the cumulative average treatment effect among noncore counties up to event time s:

$$\delta_s^{nc} = \frac{1}{N_{nc}} \sum_{i:i\in nc} \sum_{t:A_i \le t \le A_i+s} \delta_{it} .$$

In the placebo study, the true treatment effect is zero by construction. Therefore, performance is evaluated by comparing aggregated predicted untreated outcomes to their observed untreated counterparts.

For each replication r, subgroup g (e.g., noncore), and event time s, define the cumulative observed untreated outcome

$$\bar{Y}_{g,\le s}^{(r)} = \frac{1}{N_g} \sum_{i:i\in g} \sum_{t:A_i^{(r)} \le t \le A_i^{(r)}+s} Y_{it}\,(0),$$

and the corresponding cumulative predicted untreated outcome

$$\bar{\hat{Y}}_{g,\le s}^{(r)} = \frac{1}{N_g} \sum_{i:i\in g} \sum_{t:A_i^{(r)} \le t \le A_i^{(r)}+s} \hat{Y}_{it}^{(r)} .$$

Performance measures are computed using these aggregated quantities.

Point Prediction Measures

For each replication r, subgroup g, and event time s, define:

- Level-scale error

$$E_{g,s}^{(r)} = \bar{Y}_{g,\le s}^{(r)} - \bar{\hat{Y}}_{g,\le s}^{(r)}$$

- Relative error

$$RD_{g,s}^{(r)} = \frac{E_{g,s}^{(r)}}{\bar{\hat{Y}}_{g,\le s}^{(r)}},$$

These quantities are summarized across replications by their mean (bias), the mean of their absolute value (mean absolute error, MAE), as well as the standard deviations. The root mean square error is defined as,

$$\mathrm{RMSE}_{g,s} = \sqrt{\frac{1}{R}\sum_{r=1}^{R}\left(E_{g,s}^{(r)}\right)^2} .$$

Distributional scoring measures

Posterior predictive intervals are constructed by aggregating posterior draws first and then computing quantiles of the aggregated distribution. Specifically, for each posterior draw b, we compute the aggregated cumulative prediction

$$\bar{\hat{Y}}_{g,\le s}^{(r,b)},$$

and obtain interval bounds from the empirical quantiles across b.

- Coverage

$$\mathrm{Coverage}_{g,s}^{(r)} = \mathbf{1}\left\{ \hat{l}_{g,s}^{(r)} \le \bar{Y}_{g,\le s}^{(r)} \le \hat{u}_{g,s}^{(r)} \right\}.$$

- Interval score

For nominal level $1-\alpha$,

$$IS_{g,s}^{(r)}(\alpha) = (\hat{u}_{g,s}^{(r)} - \hat{l}_{g,s}^{(r)}) + \frac{2}{\alpha}(\hat{l}_{g,s}^{(r)} - \bar{Y}_{g,\leq s}^{(r)})\mathbf{1}\{\bar{Y}_{g,\leq s}^{(r)} < \hat{l}_{g,s}^{(r)}\} + \frac{2}{\alpha}(\bar{Y}_{g,\leq s}^{(r)} - \hat{u}_{g,s}^{(r)})\mathbf{1}\{\bar{Y}_{g,\leq s}^{(r)} > \hat{u}_{g,s}^{(r)}\}.$$

- Continuous ranked probability score (CRPS)

To evaluate the full predictive distribution without requiring an explicit likelihood for the aggregated quantity, we compute the CRPS using posterior samples. Let

$$x_{g,s}^{(r)} = \left\{\bar{\tilde{Y}}_{g,\leq s}^{(r,b)}\right\}_{b=1}^{B} \text{ and } \quad y_{g,s}^{(r)} = \bar{Y}_{g,\leq s}^{(r)}.$$

The sample-based CRPS is

$$CRPS_{g,s}^{(r)} = \frac{1}{B}\sum_{b=1}^{B} | x_{g,s,b}^{(r)} - y_{g,s}^{(r)} | \; - \; \frac{1}{2B^2}\sum_{b=1}^{B}\sum_{b'=1}^{B} | x_{g,s,b}^{(r)} - x_{g,s,b'}^{(r)} |.$$

This score is proper and summarizes both calibration and sharpness of the predictive distribution.

Interpretation

All performance measures are computed at the same aggregation level as the estimand in Section 2.2. This ensures that evaluation targets the ability of each method to recover subgroup-specific cumulative effects, rather than individual unit–year counterfactuals.

Because no treatment effect is introduced in the placebo simulations, the true cumulative effect is zero in all worlds. Deviations from zero therefore reflect predictive error in estimating untreated outcomes under alternative assignment mechanisms.

2. Additional results for the placebo study

This section presents additional performance results beyond those shown in the main text. Specifically, we report results for core counties and for the full sample (i.e., without restricting to the trusted subset), alongside the corresponding summary table of performance measures.

Across data-generating processes, the trusted-sample results confirm the patterns observed in the main text: model rankings based on aggregated absolute error are broadly consistent, and differences across approaches are modest at event time 0 but become more pronounced at longer horizons. When the full sample is considered, overall error increases due to the inclusion of counties with poor pre-treatment predictive fit. However, comparative model performance remains qualitatively similar. In core counties, performance differences are generally smaller than in noncore counties, and in scenarios aligned with the latent-level data-generating process, TWFE performs competitively, consistent with its additive fixed-effect structure.

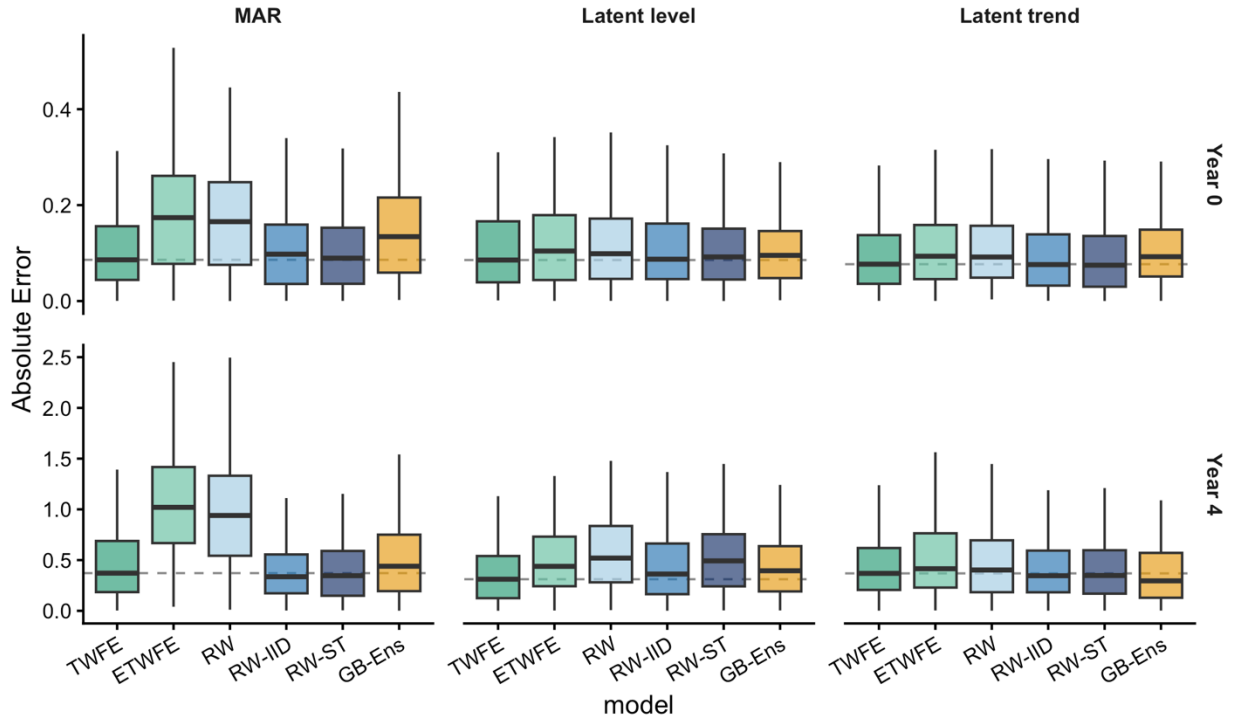


Figure A1 Predictive performance in placebo simulations among core counties, reliability-restricted sample (excluding the top 10% of counties by pre-period RMSE). Panels correspond to different data-generating processes (MAR, latent level, latent trend) and post-intervention horizons (year 0 and year 4). Performance is measured by root mean squared error (RMSE) of the estimated cumulative effect. Lower values indicate better predictive accuracy. Models shown are: two-way fixed effects (TWFE), enhanced two-way fixed effects (ETWFE), random walk (RW), random walk with independent and identically distributed county effects (RW-

IID), random walk with spatiotemporal structure (RW-ST), and gradient boosting ensemble (GB-Ens). Boxes summarize performance across placebo replications.

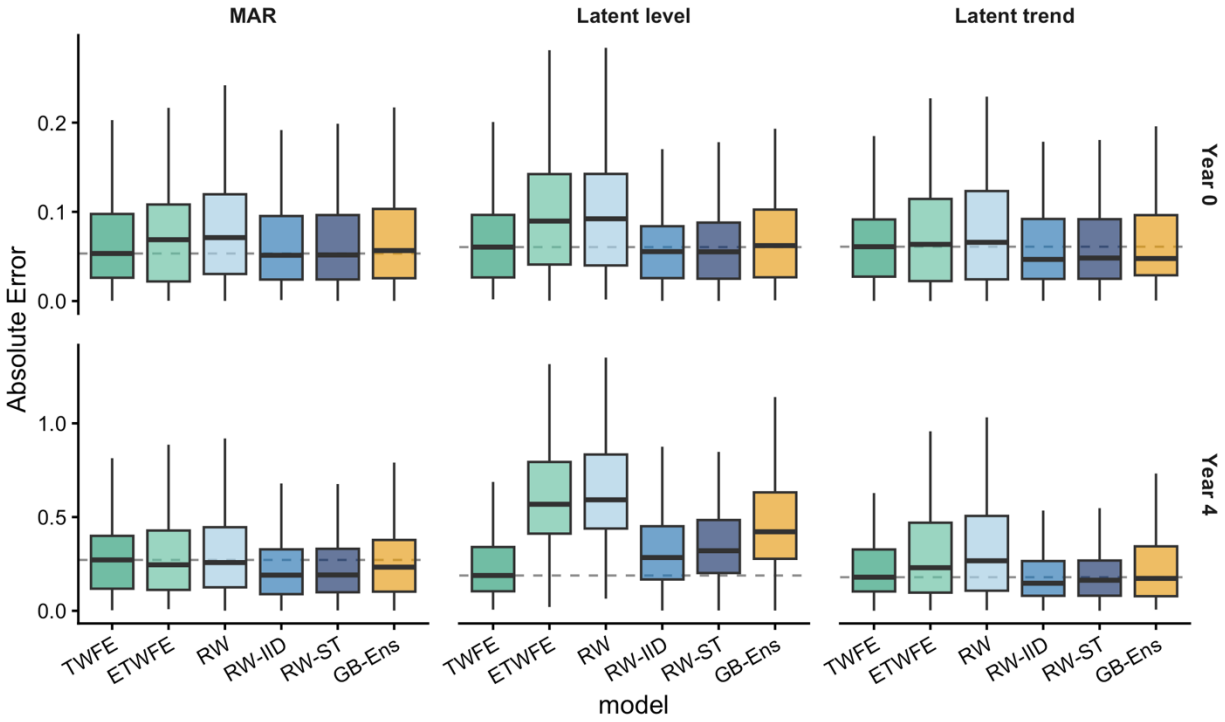


Figure A2 Predictive performance in placebo simulations among noncore counties, full sample. *Panels correspond to different data-generating processes (MAR, latent level, latent trend) and post-intervention horizons (year 0 and year 4). Performance is measured by root mean squared error (RMSE) of the estimated cumulative effect. Lower values indicate better predictive accuracy. Models shown are: two-way fixed effects (TWFE), enhanced two-way fixed effects (ETWFE), random walk (RW), random walk with independent and identically distributed county effects (RW-IID), random walk with spatiotemporal structure (RW-ST), and gradient boosting ensemble (GB-Ens). Boxes summarize performance across placebo replications.*

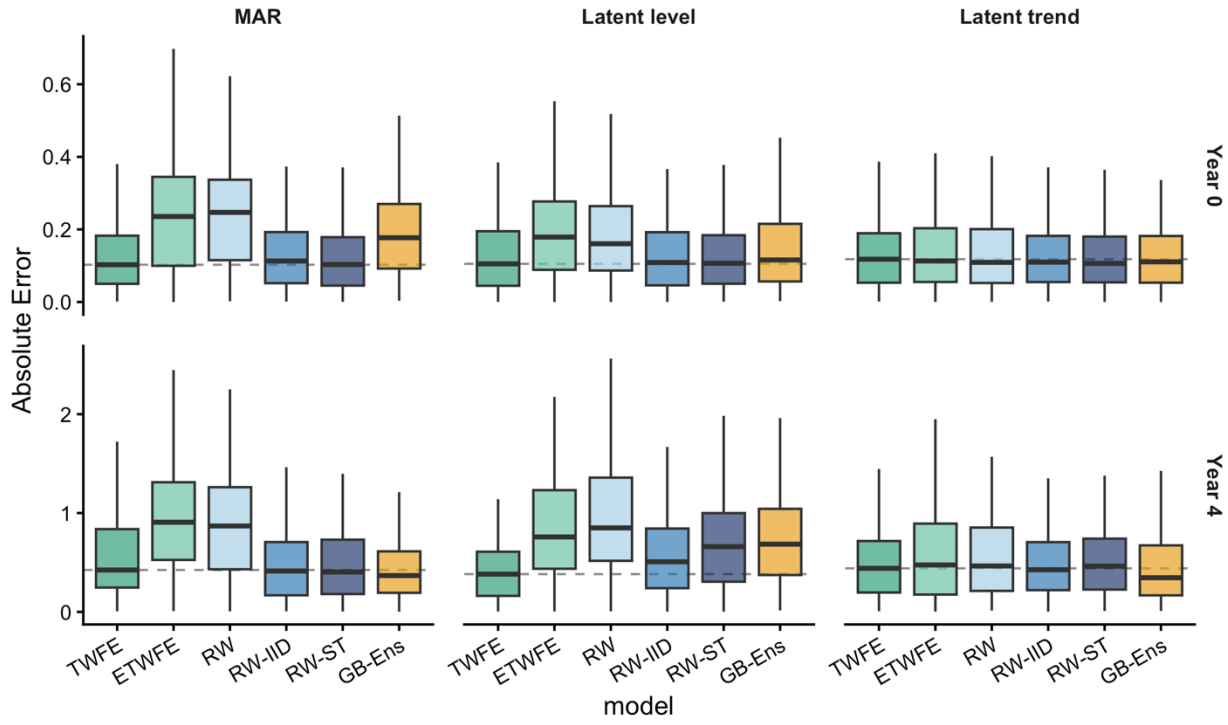


Figure A3 Predictive performance in placebo simulations among core counties, full sample*. Panels correspond to different data-generating processes (MAR, latent level, latent trend) and post-intervention horizons (year 0 and year 4). Performance is measured by root mean squared error (RMSE) of the estimated cumulative effect. Lower values indicate better predictive accuracy. Models shown are: two-way fixed effects (TWFE), enhanced two-way fixed effects (ETWFE), random walk (RW), random walk with independent and identically distributed county effects (RW-IID), random walk with spatiotemporal structure (RW-ST), and gradient boosting ensemble (GB-Ens). Boxes summarize performance across placebo replications.*

Model	Years	Bias (SD)	MAE (SD)	RMSE	CRPS	Coverage	Interval score	Relative	
								Bias (SD)	MAE (SD)
MAR									
TWFE	0	−0.004 (0.074)	0.057 (0.046)	0.071	0.041	0.940	0.378	−0.014 (0.267)	0.208 (0.167)
ETWFE	0	0.004 (0.072)	0.055 (0.046)	0.071	0.040	0.950	0.368	0.016 (0.272)	0.209 (0.175)
RW	0	0.017 (0.071)	0.056 (0.046)	0.071	0.041	0.935	0.378	0.066 (0.283)	0.222 (0.186)
RW-IID	0	0.013 (0.071)	0.055 (0.047)	0.071	0.040	0.935	0.368	0.050 (0.277)	0.214 (0.182)
RW-ST	0	0.011 (0.071)	0.055 (0.046)	0.071	0.040	0.930	0.394	0.041 (0.274)	0.211 (0.178)
GB-Ens	0	−0.003 (0.071)	0.055 (0.046)	0.071	0.040	0.935	0.367	−0.011 (0.263)	0.202 (0.169)
TWFE	4	−0.096 (0.252)	0.214 (0.164)	0.268	0.152	0.900	1.350	−0.061 (0.173)	0.147 (0.110)
ETWFE	4	−0.067 (0.212)	0.182 (0.127)	0.221	0.127	0.955	0.988	−0.045 (0.152)	0.129 (0.092)
RW	4	−0.013 (0.209)	0.170 (0.121)	0.207	0.120	0.955	0.963	−0.008 (0.154)	0.125 (0.090)
RW-IID	4	−0.009 (0.217)	0.177 (0.126)	0.217	0.123	0.970	1.035	−0.006 (0.161)	0.130 (0.094)
RW-ST	4	−0.021 (0.216)	0.176 (0.126)	0.217	0.124	0.980	1.022	−0.015 (0.158)	0.129 (0.093)
GB-Ens	4	−0.092 (0.206)	0.183 (0.131)	0.226	0.128	0.950	1.052	−0.064 (0.144)	0.127 (0.092)
Latent level									
TWFE	0	−0.015 (0.058)	0.048 (0.036)	0.063	0.034	0.950	0.283	−0.052 (0.213)	0.176 (0.130)
ETWFE	0	0.053 (0.056)	0.063 (0.044)	0.077	0.046	0.820	0.440	0.262 (0.276)	0.311 (0.219)
RW	0	0.056 (0.056)	0.065 (0.044)	0.077	0.048	0.800	0.453	0.277 (0.276)	0.323 (0.220)

Model	Years	Bias (SD)	MAE (SD)	RMSE	CRPS	Coverage	Interval score	Relative	
								Bias (SD)	MAE (SD)
RW-IID	0	0.029 (0.057)	0.051 (0.038)	0.063	0.037	0.920	0.292	0.129 (0.248)	0.225 (0.165)
RW-ST	0	0.029 (0.056)	0.051 (0.037)	0.063	0.037	0.920	0.292	0.131 (0.247)	0.227 (0.164)
GB-Ens	0	0.025 (0.057)	0.050 (0.036)	0.063	0.036	0.930	0.276	0.108 (0.243)	0.215 (0.156)
TWFE	4	−0.203 (0.193)	0.228 (0.162)	0.279	0.164	0.780	1.700	−0.137 (0.129)	0.155 (0.106)
ETWFE	4	0.236 (0.155)	0.244 (0.143)	0.283	0.178	0.680	2.004	0.241 (0.162)	0.248 (0.151)
RW	4	0.266 (0.156)	0.272 (0.147)	0.308	0.207	0.600	2.723	0.279 (0.168)	0.284 (0.159)
RW-IID	4	0.118 (0.168)	0.165 (0.121)	0.205	0.118	0.925	0.955	0.108 (0.152)	0.149 (0.112)
RW-ST	4	0.135 (0.165)	0.171 (0.127)	0.212	0.124	0.905	1.015	0.126 (0.153)	0.158 (0.119)
GB-Ens	4	0.115 (0.154)	0.155 (0.114)	0.192	0.111	0.930	0.863	0.104 (0.139)	0.139 (0.104)
Latent trend									
TWFE	0	−0.002 (0.063)	0.052 (0.035)	0.063	0.036	0.960	0.264	−0.003 (0.249)	0.207 (0.138)
ETWFE	0	0.026 (0.060)	0.053 (0.038)	0.063	0.038	0.925	0.282	0.117 (0.271)	0.240 (0.172)
RW	0	0.031 (0.060)	0.054 (0.041)	0.071	0.039	0.900	0.321	0.145 (0.277)	0.250 (0.188)
RW-IID	0	0.020 (0.061)	0.052 (0.037)	0.063	0.037	0.930	0.284	0.090 (0.265)	0.228 (0.162)
RW-ST	0	0.019 (0.060)	0.051 (0.036)	0.063	0.036	0.945	0.272	0.083 (0.261)	0.223 (0.158)
GB-Ens	0	0.003 (0.061)	0.051 (0.033)	0.063	0.035	0.945	0.262	0.015 (0.247)	0.206 (0.136)
TWFE	4	−0.082 (0.182)	0.162 (0.115)	0.197	0.112	0.950	0.854	−0.059 (0.140)	0.124 (0.087)

Model	Years	Bias (SD)	MAE (SD)	RMSE	CRPS	Coverage	Interval score	Relative	
								Bias (SD)	MAE (SD)
ETWFE	4	0.090 (0.157)	0.145 (0.108)	0.182	0.102	0.935	0.882	0.084 (0.142)	0.131 (0.100)
RW	4	0.129 (0.156)	0.164 (0.119)	0.202	0.117	0.910	0.978	0.123 (0.146)	0.153 (0.114)
RW-IID	4	0.072 (0.162)	0.143 (0.104)	0.176	0.101	0.960	0.815	0.066 (0.143)	0.126 (0.094)
RW-ST	4	0.070 (0.159)	0.140 (0.102)	0.173	0.100	0.975	0.786	0.064 (0.139)	0.124 (0.091)
GB-Ens	4	−0.004 (0.155)	0.124 (0.094)	0.155	0.088	0.980	0.720	−0.001 (0.128)	0.102 (0.077)

Table A 1 Performance measures from placebo simulations for cumulative untreated outcomes among noncore counties at event times 0 and 4, restricted to the trusted sample (counties excluding the upper 10% of pre-treatment prediction error within model and replication). Results are shown separately for the three data-generating processes (MAR, latent level, latent trend). Bias and MAE denote the mean and mean absolute error across replications (standard deviations in parentheses). RMSE is the square root of the mean squared error across replications. CRPS and interval score summarize the posterior predictive distribution of the aggregated outcome, and Coverage denotes the proportion of replications in which the nominal 95% posterior interval contains the observed aggregated outcome. Relative measures report percent deviation from the predicted aggregated outcome.

Model	Years	Bias (SD)	MAE (SD)	RMSE	CRPS	Coverage	Interval score	Relative	
								Bias (SD)	MAE (SD)
MAR									
TWFE	0	−0.041 (0.124)	0.105 (0.077)	0.130	0.075	0.935	0.591	−0.033 (0.105)	0.089 (0.064)
ETWFE	0	−0.170 (0.131)	0.180 (0.117)	0.214	0.137	0.720	1.408	−0.128 (0.097)	0.136 (0.085)
RW	0	−0.154 (0.127)	0.167 (0.110)	0.200	0.121	0.760	1.288	−0.117 (0.095)	0.127 (0.082)
RW-IID	0	−0.069 (0.117)	0.108 (0.082)	0.134	0.077	0.890	0.621	−0.056 (0.096)	0.089 (0.067)

Model	Years	Bias (SD)	MAE (SD)	RMSE	CRPS	Coverage	Interval score	Relative	
								Bias (SD)	MAE (SD)
RW-ST	0	−0.059 (0.117)	0.104 (0.079)	0.130	0.074	0.925	0.595	−0.048 (0.097)	0.087 (0.065)
GB-Ens	0	−0.128 (0.120)	0.145 (0.099)	0.176	0.101	0.845	0.884	−0.100 (0.093)	0.113 (0.076)
TWFE	4	0.147 (0.569)	0.465 (0.358)	0.587	0.335	0.905	2.883	0.028 (0.094)	0.077 (0.061)
ETWFE	4	−1.061 (0.577)	1.073 (0.554)	1.207	0.815	0.565	9.779	−0.142 (0.070)	0.144 (0.066)
RW	4	−0.983 (0.620)	1.001 (0.590)	1.161	0.745	0.555	10.250	−0.132 (0.076)	0.135 (0.070)
RW-IID	4	0.032 (0.506)	0.398 (0.313)	0.506	0.284	0.980	2.499	0.007 (0.080)	0.064 (0.049)
RW-ST	4	0.091 (0.503)	0.403 (0.312)	0.510	0.291	0.970	2.429	0.017 (0.081)	0.066 (0.050)
GB-Ens	4	−0.423 (0.475)	0.509 (0.381)	0.636	0.346	0.940	2.996	−0.062 (0.069)	0.075 (0.054)
Latent level									
TWFE	0	0.001 (0.137)	0.106 (0.086)	0.138	0.077	0.900	0.670	0.004 (0.137)	0.107 (0.085)
ETWFE	0	0.086 (0.130)	0.123 (0.097)	0.155	0.087	0.870	0.795	0.099 (0.147)	0.138 (0.111)
RW	0	0.084 (0.128)	0.120 (0.096)	0.155	0.088	0.850	0.891	0.096 (0.144)	0.134 (0.109)
RW-IID	0	0.032 (0.131)	0.107 (0.081)	0.134	0.076	0.915	0.647	0.035 (0.135)	0.112 (0.083)
RW-ST	0	0.049 (0.128)	0.109 (0.083)	0.138	0.078	0.910	0.666	0.053 (0.135)	0.116 (0.087)
GB-Ens	0	0.039 (0.129)	0.107 (0.081)	0.134	0.076	0.950	0.657	0.043 (0.135)	0.114 (0.085)
TWFE	4	−0.027 (0.461)	0.364 (0.283)	0.462	0.262	0.935	2.377	−0.004 (0.094)	0.074 (0.058)
ETWFE	4	0.452 (0.484)	0.534 (0.391)	0.662	0.373	0.790	4.049	0.108 (0.114)	0.124 (0.096)

Model	Years	Bias (SD)	MAE (SD)	RMSE	CRPS	Coverage	Interval score	Relative	
								Bias (SD)	MAE (SD)
RW	4	0.530 (0.495)	0.602 (0.404)	0.724	0.450	0.695	5.956	0.128 (0.118)	0.142 (0.101)
RW-IID	4	0.351 (0.435)	0.442 (0.341)	0.558	0.323	0.910	2.556	0.077 (0.095)	0.097 (0.075)
RW-ST	4	0.484 (0.412)	0.525 (0.358)	0.635	0.378	0.870	3.039	0.109 (0.092)	0.118 (0.080)
GB-Ens	4	0.390 (0.417)	0.457 (0.342)	0.570	0.334	0.910	2.793	0.088 (0.093)	0.102 (0.077)
Latent trend									
TWFE	0	0.009 (0.121)	0.095 (0.075)	0.122	0.068	0.930	0.559	0.012 (0.130)	0.102 (0.080)
ETWFE	0	−0.052 (0.131)	0.113 (0.084)	0.141	0.081	0.915	0.702	−0.047 (0.129)	0.112 (0.079)
RW	0	−0.044 (0.129)	0.110 (0.081)	0.138	0.077	0.905	0.683	−0.040 (0.128)	0.109 (0.076)
RW-IID	0	−0.009 (0.117)	0.092 (0.072)	0.118	0.067	0.955	0.550	−0.008 (0.122)	0.096 (0.076)
RW-ST	0	0.002 (0.115)	0.089 (0.072)	0.114	0.065	0.955	0.541	0.004 (0.122)	0.094 (0.077)
GB-Ens	0	−0.059 (0.120)	0.109 (0.077)	0.134	0.075	0.925	0.632	−0.056 (0.118)	0.108 (0.074)
TWFE	4	0.110 (0.559)	0.455 (0.342)	0.568	0.328	0.865	3.193	0.028 (0.119)	0.097 (0.075)
ETWFE	4	−0.342 (0.559)	0.521 (0.396)	0.654	0.388	0.820	4.014	−0.058 (0.106)	0.099 (0.070)
RW	4	−0.249 (0.587)	0.498 (0.397)	0.636	0.363	0.800	3.932	−0.040 (0.114)	0.096 (0.072)
RW-IID	4	0.118 (0.506)	0.416 (0.310)	0.519	0.295	0.900	2.531	0.028 (0.108)	0.089 (0.067)
RW-ST	4	0.195 (0.494)	0.424 (0.318)	0.529	0.303	0.915	2.484	0.045 (0.108)	0.093 (0.070)
GB-Ens	4	−0.148 (0.452)	0.372 (0.296)	0.474	0.267	0.935	2.315	−0.027 (0.091)	0.075 (0.059)

Table A 2 Performance measures from placebo simulations for cumulative untreated outcomes among ncore counties at event times 0 and 4, restricted to the trusted sample (counties excluding the upper 10% of pre-treatment prediction error within model and replication). Results are shown separately for the three data-generating processes (MAR, latent level, latent trend). Bias and MAE denote the mean and mean absolute error across replications (standard deviations in parentheses). RMSE is the square root of the mean squared error across replications. CRPS and interval score summarize the posterior predictive distribution of the aggregated outcome, and Coverage denotes the proportion of replications in which the nominal 95% posterior interval contains the observed aggregated outcome. Relative measures report percent deviation from the predicted aggregated outcome.

S2. Covariate Adjustment

In the main text, all models condition exclusively on the history of the outcome process. This choice reflects our view that, in small-area settings with sparse count outcomes, sufficiently rich outcome histories often capture most of the predictable variation relevant for counterfactual imputation. Nonetheless, we have access to additional county-level characteristics—including unemployment rates, poverty levels, median household income, and the percentage of American Indian/Alaska Native residents—that are plausibly unaffected by the intervention.

Although, in our experience, these covariates provide limited additional predictive power once outcome histories are included, we examine their incorporation for completeness. This appendix describes how covariates can be introduced into each model and reports corresponding simulation results.

Model-Specific Incorporation of Covariates

TWFE

In the two-way fixed-effects specification, covariates enter additively:

$$\eta_{it} = \alpha_i + \gamma_t + \beta X_{it}.$$

To allow for greater flexibility, the coefficient vector β can be permitted to vary over time:

$$\eta_{it} = \alpha_i + \gamma_t + \beta_t X_{it}.$$

This relaxation mitigates the reliance on a strict common-trend assumption conditional on covariates.

ETWFE

Following Wooldridge (2023), the cohort-based specification allows coefficients to vary by cohort and time to further increase the tenability of common-trend assumptions:

$$\eta_{it} = \alpha_{A_i} + \gamma_t^{(A_i)} + \left(\pi + \kappa^{(A_i)} + \lambda_t\right) X_{it}.$$

where:

- π is a global effect,

- $\kappa^{(A_i)}$ captures cohort-specific deviations,
- λ_t captures time-varying deviations common across cohorts.

Hierarchical Models (RW, RW-IID, RW-ST)

For the hierarchical specifications, covariate effects are embedded within the same partial-pooling framework used for latent heterogeneity. Specifically, we assign:

$$\kappa^{(A_i)} \sim \mathcal{N}(\mu_\kappa, \tau_\kappa^{-1}),$$

and model the time component as random walks:

$$\lambda_t = \lambda_{t-1} + \varsigma_t, \varsigma_t \sim \mathcal{N}(0, \tau_\varsigma^{-1}).$$

This specification allows covariate effects to vary across cohorts and over time while inducing shrinkage toward common patterns. The degree of pooling is governed by the corresponding precision parameters.

Gradient Boosting

For the gradient boosting specification, covariates are included directly in the feature vector and treated identically to other predictors derived from outcome histories. No additional structural assumptions are imposed; nonlinearities and interactions are learned adaptively through the boosting procedure.

Simulation Results with Covariates

Incorporating additional covariates does not materially alter predictive performance across models in the reliability-restricted sample. Performance rankings remain broadly unchanged. Modest improvements are observed for ETWFE and RW under the latent-level assignment mechanism at longer horizons (year 4). These specifications do not include unit-specific fixed effects, and covariates may partially capture persistent cross-county heterogeneity in this setting. In contrast, models that already include unit-level effects show little sensitivity to covariate adjustment.

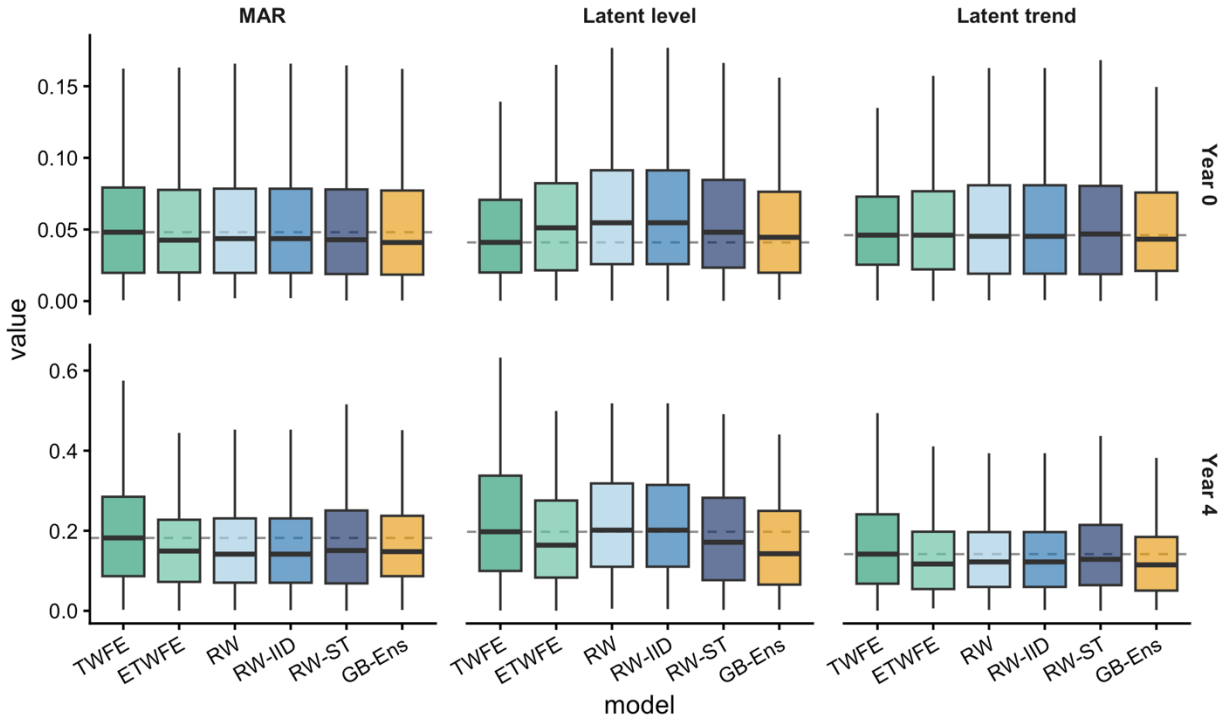


Figure A4 Predictive performance in placebo simulations when including covariates among noncore counties, reliability-restricted sample (excluding the top 10% of counties by pre-period RMSE). Panels correspond to different data-generating processes (MAR, latent level, latent trend) and post-intervention horizons (year 0 and year 4). Performance is measured by root mean squared error (RMSE) of the estimated cumulative effect. Lower values indicate better predictive accuracy. Models shown are: two-way fixed effects (TWFE), enhanced two-way fixed effects (ETWFE), random walk (RW), random walk with independent and identically distributed county effects (RW-IID), random walk with spatiotemporal structure (RW-ST), and gradient boosting ensemble (GB-Ens). Boxes summarize performance across placebo replications.

S3. Additional Case Study Results

This appendix provides supplementary analysis of effect heterogeneity in the case study. In particular, we examine whether county-level variation in estimated cumulative effects through year 4 is associated with the proportion of long trainings implemented within each county.

	Full sample			**Reliability-restricted** (excludes top 10% pre-fit RMSE)		
		95% Credible Interval			95% Credible Interval	
Model	Estimate	Lower limit	Upper limit	Estimate	Lower limit	Upper limit
ETWFE	0.206	0.072	0.332	0.075	−0.077	0.218
RW	0.178	0.062	0.298	0.046	−0.082	0.178
RW-IID	0.084	−0.088	0.215	−0.009	−0.167	0.144
RW-ST	0.086	−0.068	0.237	−0.025	−0.188	0.148
TWFE	0.005	−0.145	0.166	−0.030	−0.193	0.146
GB-Ens	0.117	−0.022	0.263	−0.010	−0.181	0.147

Table A 3 Spearman rank correlation between the estimated county-level cumulative effect (through year 4) and the proportion of long trainings implemented within each county. Correlations are computed separately by model. For each posterior draw, the county-specific cumulative effect is aggregated through event time 4, and the Spearman correlation with the proportion of long trainings is calculated; reported values summarize the posterior distribution of this correlation. Credible intervals reflect posterior uncertainty. Across models, estimated correlations are generally small and include zero, indicating limited evidence of association between training composition and estimated effect size.